

The cytotoxicity of protoneodioscin (NSC-698789), a furostanol saponin from the rhizomes of *Dioscorea collettii* var. *hypoglauca*, against human cancer cells *in vitro*

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Summary

Protoneodioscin (NSC-698789) is a furostanol saponin isolated from the rhizomes of *Dioscorea collettii* var. *hypoglauca* (Dioscoreaceae), a Chinese herbal remedy for cancer treatment. Our studies showed that protoneodioscin is cytotoxic against most cell lines from leukemia and solid tumors in the NCI's (National Cancer Institute, USA) anticancer drug screen. Leukemia, CNS cancer, and prostate cancer are the most sensitive subpanels to protoneodioscin, while melanoma, ovarian cancer, and renal cancer are less sensitive. The preliminary animal studies showed that the maximum tolerant dose of protoneodioscin was 600 mg/kg to mice. Based on an analysis of the COMPARE software with protoneodioscin as a seed compound, no compounds in the NCI's database have similar cytotoxicity patterns to those of protoneodioscin, indicating a potentially novel mechanism of anticancer action involved.

Key words: Cytotoxicity, protoneodioscin, NSC-698789, *Dioscorea collettii* var. *hypoglauca*, Dioscoreaceae

■ Introduction

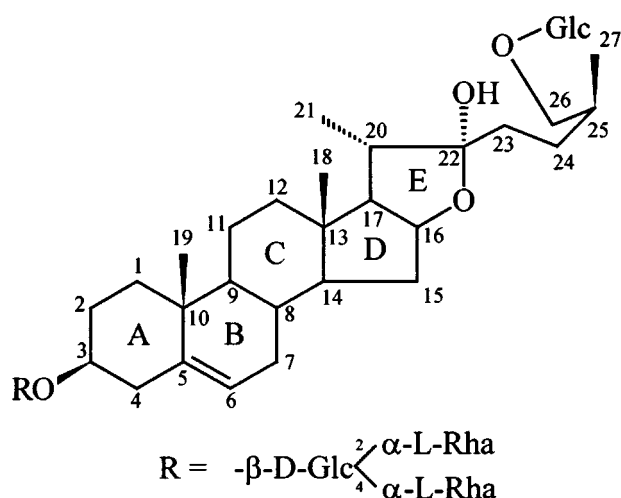
Dioscorea collettii var. *hypoglauca* (Dioscoreaceae) is a traditional Chinese herbal remedy used in the treatment of cervical carcinoma, carcinoma of urinary bladder, and renal tumor. It was included in The Pharmacopoeia of People's Republic of China (1985, 1990, 1995, and 2000 versions). By bioactivity-guided isolation, fourteen steroidal saponins were isolated from the rhizome of *D. collettii* var. *hypoglauca* and showed antifungal activity inducing morphological deformation of mycelia and conidia of *Pyricularia oryzae* (a phytopathogenic fungus responsible for rice blast) (Kobayashi et al. 1996), including three spirostanol saponins (Hu et al. 1996), nine furostanol saponins (Hu et al. 1997a, b), and two pregnane glycosides (Hu et al. 1998, 1999).

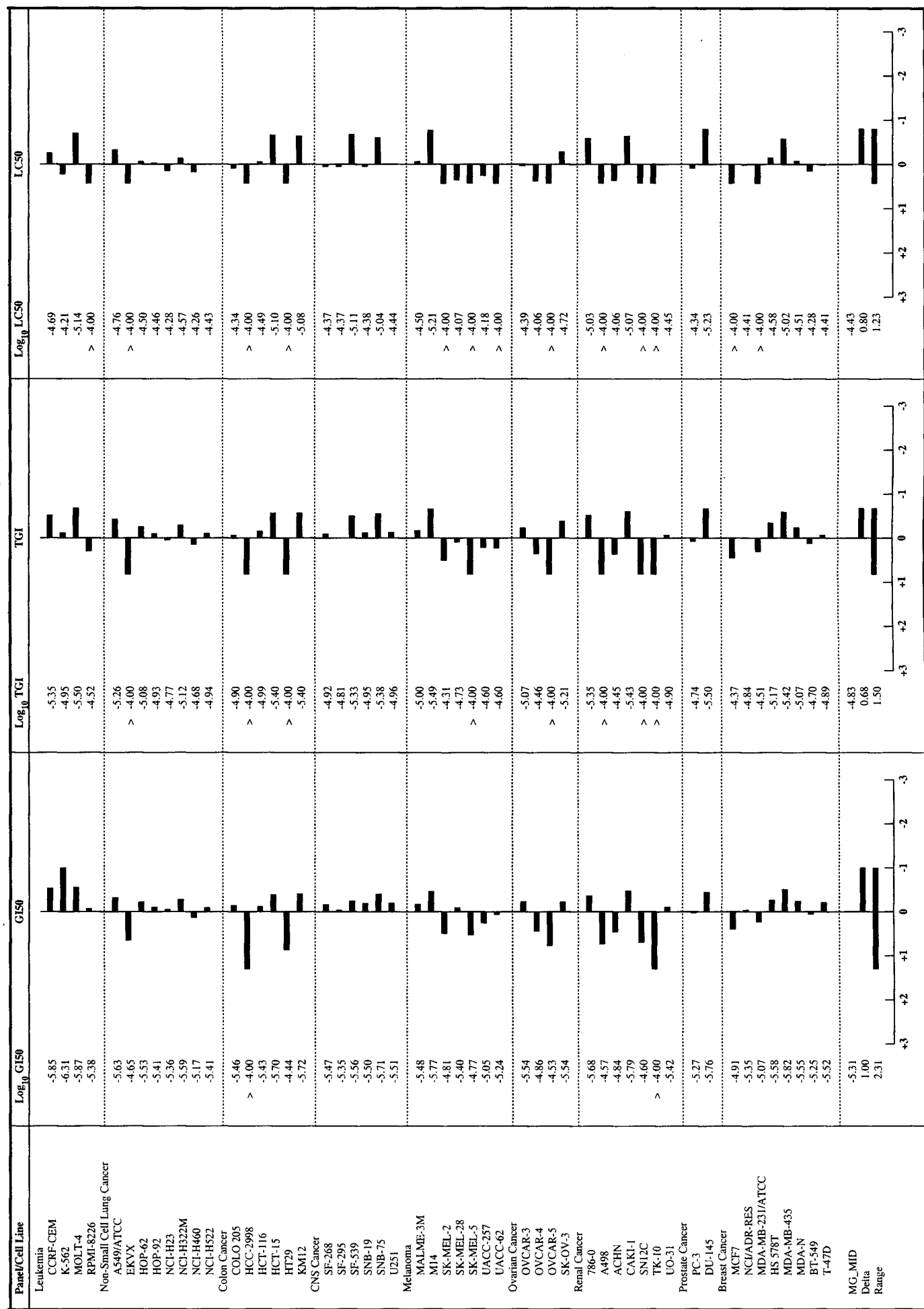
The preliminary cytotoxicity studies demonstrated that eleven of fourteen steroidal saponins were active against the human acute myeloid leukemia (AML) K562 cells *in vitro* (Hu et al. 1996, 1997a). Recently, a systemic study of methyl protogracillin (NSC-698792) was reported using the NCI's (National Cancer Institute, USA) anticancer drug screen (Hu and Yao, 2001), an *in vitro* disease-oriented screening system with a panel of 60 human cancer cell lines from leukemia, non-small cell lung cancer (NSCLC), colon cancer, central nervous system (CNS) cancer, prostate cancer, melanoma, ovarian cancer, renal cancer, and breast cancer (Boyd, 1989; Monks et al. 1991; Grever et al. 1992). As our continuous studies, this paper presents the cytotoxicity of another furostanol saponin, proto-

Table 1. The cytotoxicity of protoneodioscin (NSC-698789) against human cancer cells *in vitro*.

Panel/Cell Line	GI ₅₀ (μ M)	TGI (μ M)	TGI/ GI ₅₀	LC ₅₀ (μ M)	LC ₅₀ / GI ₅₀	Panel/Cell Line	GI ₅₀ (μ M)	TGI (μ M)	TGI/ GI ₅₀	LC ₅₀ (μ M)	LC ₅₀ / GI ₅₀
Leukemia						Melanoma					
CCRF-CEM	1.42	4.47	3.15	20.4	14.4	MALME-3M	3.28	9.97	3.04	31.6	9.63
K-562	0.49	11.2	22.9	61.4	125	M14	1.70	3.24	1.91	6.18	3.64
MOLT-4	1.36	3.13	2.30	7.22	5.31	SK-MEL-2	15.4	49.0	3.18	>100	>6.49
RPMT-8226	4.16	30.1	7.24	>100	>2.40	SK-MEL-28	3.96	18.6	4.70	84.2	21.3
NSCLC						SK-MEL-5	16.9	>100	>5.92	>100	>5.92
A549/ATCC	2.35	5.55	2.36	17.3	7.36	UACC-257	8.95	24.9	2.78	66.4	7.42
EKVX	22.3	>100	>4.48	>100	>4.48	UACC-62	5.71	25.4	4.45	>100	>17.5
HOP-62	2.95	8.25	2.80	31.5	>10.7	Ovarian Cancer					
HOP-92	3.85	11.8	3.06	34.3	8.91	OVCAR-3	2.87	8.59	2.99	40.3	14.0
NCI-H23	4.34	16.9	3.89	52.7	12.1	OVCAR-4	13.7	34.7	2.53	87.9	6.42
NCI-H322M	2.55	7.58	2.97	26.7	10.5	OVCAR-5	29.4	>100	>3.40	>100	>3.40
NCI-H460	6.74	20.9	3.10	55.0	8.16	SK-OV-3	2.91	6.11	2.10	19.2	6.60
NCI-H522	3.88	11.6	2.99	37.5	9.66	Renal Cancer					
Colon Cancer						786-0	2.11	4.45	2.11	9.38	4.45
COLO 205	3.48	12.6	3.62	45.9	13.2	A498	26.7	>100	>3.75	>100	3.75
HCC-2998	>100	>100	ND	>100	ND*	ACHN	14.3	35.2	2.46	86.4	6.04
HCT-116	3.69	10.3	2.79	32.1	8.70	CAKI-1	1.62	3.69	2.28	8.42	5.20
HCT-15	2.00	3.98	1.99	7.94	3.97	SN12C	25.0	>100	>4.00	>100	>4.00
HT29	36.7	>100	>2.72	>100	>2.72	TK-10	>100	>100	ND	>100	ND
KM12	1.92	4.01	2.09	8.37	4.36	UO-31	3.81	12.6	3.31	35.4	9.29
CNS Cancer						Breast Cancer					
SF-268	3.35	12.0	3.58	42.2	12.6	MCF-7	12.3	42.9	3.49	>100	>8.13
SF-295	4.51	15.4	3.41	42.4	9.40	NCI/ADR-RES	4.44	14.4	3.24	39.2	8.83
SF-539	2.78	4.65	1.67	7.77	2.79	MDA-MB-231	8.55	30.8	3.60	>100	>11.7
SNB-19	3.14	11.2	3.57	41.6	13.2	HS 578T	2.63	6.71	2.55	26.1	9.92
SNB-75	1.93	4.18	2.17	9.03	4.68	MDA-MB-435	1.50	3.81	2.54	9.65	6.43
U251	3.11	10.9	3.50	36.7	11.8	MDA-N	2.80	8.49	3.03	30.9	11.0
Prostate Cancer						BT-549	5.62	20.0	3.56	52.7	9.38
PC-3	5.32	18.3	3.44	45.9	8.63	T-47D	3.01	12.8	4.25	38.9	12.9
DU-145	1.74	3.18	1.83	5.83	3.35						

*ND: not determined

**Fig. 1.** The structure of protoneodioscin (NSC-698789).



neodioscin (NSC-698789) (Fig. 1), against human cancer cell lines *in vitro*, as well as toxicity studies in mice.

Materials and Methods

Chemical

Protoneodioscin (NSC-698789) was isolated from the rhizome of *D. collettii* var. *hypoglauc*a and its physicochemical properties and spectral data were reported in our previous studies (Hu et al. 1997a). The stock solutions were prepared in dimethyl sulfoxide (DMSO) at 10 mM and stored at -20°C .

The NCI's Anticancer Drug Screen

A total of 60 cell lines were from human leukemia, melanoma, and cancers of lung, colon, brain, ovary, renal, breast, and prostate. The cell lines were adaptable to a single growth medium and had reproducible profiles for growth and drug sensitivity (Alley et al. 1988; Boyd, 1989).

The assay and data analysis were followed the same procedure in our previous report (Hu and Yao, 2001). In brief, each cell line was suspended in the medium and inoculated onto 96-well microplates on day zero. After 24-h incubation, the stock solution of protoneodioscin in DMSO was diluted to five concentrations (i.e. 10^{-4} , 10^{-5} , 10^{-6} , 10^{-7} , and 10^{-8} M) and incubated with cells for further 48 h. The cells were fixed *in situ*, washed and dried. Sulforhodamine B (SRB), a protein stain binding to basic amino acids of cellular macromolecules, was added, washed, and dried (Rubinstein et al. 1990; Skehan et al. 1990). The bound stain was solubilized with Tris buffer, and the optical density (OD) was measured spectrophotometrically at 515 nm on an automatic plate reader interfaced with a microcomputer. Each concentration was tested with three replications.

The dose response curves were plotted by the percentage of growth (PG) against the $\log_{10}$ of drug concentration. The drug concentrations, where the PG horizontal lines at +50, 0 and -50 levels crossed dose response curves, represented the interpolated values to cause 50% growth inhibition (GI_{50}), total growth inhibition (TGI), and 50% cell killing (LC_{50}), respectively.

However, not always the response parameters (GI_{50} , TGI, and LC_{50}) could be derived by interpolation. In such a case, the response parameter was given a ">" sign representing the value more than the highest tested concentration (10^{-4} M). For protoneodioscin, the calculated response parameters higher than the highest test concentration were GI_{50} against HCC-2998 (colon can-

cer) and TK-10 (renal cancer); TGI against EKVX (NSCLC), HCC-2998, HT29 (colon cancer), SK-MEL-5 (melanoma), OVCAR-5 (ovarian cancer), A498 (renal cancer), SN12C (renal cancer), and TK-10; LC_{50} against PRMT-8226 (leukemia), EKVX, HCC-2998, HT29, SK-MEL-2 (melanoma), SK-MEL-5, UACC-62 (melanoma), OVCAR-5, A498, SN12C, TK-10, MCF-7 (breast cancer), and MDA-MB-231 (breast cancer).

Results and Discussion

As shown in Table 1, protoneodioscin (NSC-698789) exhibited significant cytotoxicity with GI_{50} ranging from 0.5 to 9 mM against all cell lines from leukemia and most solid tumors. Very strong effects are observed against three leukemia lines (CCRF-CEM, K-562 and MOLT-4), two colon cancer lines (HCT-15 and KM12), one CNS cancer line (SNB-75), one melanoma line (M14), one renal cancer line (CAKI-1), one prostate cancer (DU-145), and one breast cancer line (MDA-MB-435) with $\text{GI}_{50} \leq 2.0 \mu\text{M}$ (i.e. 1.42, 0.49, 1.36, 2.00, 1.92, 1.93, 1.70, 1.62, 1.74 and $1.50 \mu\text{M}$). Especially, GI_{50} against the most sensitive cell line (K-562) is at sub-micromolar level at 0.49 mM. The moderate cytotoxicity was found against one NSCLC line (EKVX), one colon cancer line (HT29), two melanoma lines (SK-MEL-2 and SK-MEL-5), two ovarian cancer lines (OVCAR-4 and OVCAR-5), three renal cancer lines (A498, ACHN and SN12C), and one breast cancer line (MCF-7) with GI_{50} ranging from 10 to $40 \mu\text{M}$ (22.3, 36.7, 15.4, 16.9, 13.7, 29.4, 26.7, 14.3, 25.0 and $12.3 \mu\text{M}$). No activity was against one colon cancer line (HCC-2998) and one renal cancer line (TK-10) with $\text{GI}_{50} > 100 \text{ mM}$. The TGI values are in the 3 to 50 mM range against all sensitive cell lines ($\text{GI}_{50} < 100 \text{ mM}$) except EKVX, HT29, SK-MEL-5, OVCAR-5, A498 and SN12C with TGI $> 100 \mu\text{M}$, which are from 2- to 7-fold more than the corresponding GI_{50} values. The biggest ratio of LC_{50} to GI_{50} is found in leukemia K-562 ($\text{LC}_{50}/\text{GI}_{50} = 125$), while others are from 3- to 30-fold.

The data of response parameters in Table 1 can be plotted by the mean graphs (Fig. 2), which facilitate visual scanning of data for potential patterns of selectivity for particular cell lines or for particular subpanels with respect to a selected response parameter (Paull et al. 1989). At the GI_{50} level, leukemia, CNS cancer and prostate cancer are the most sensitive subpanels, as evidence by bars extending to the right. In contrast, melanoma, ovarian cancer, and renal cancer comprise less sensitive subpanel because at least one third of the

◀ Fig. 2. The mean graphs of protoneodioscin (NSC-698789) against human cancer cells *in vitro*.

cell lines have bars extending to the left. The bars of those $\log GI_{50} \leq -5.70$ (i.e. $GI_{50} \leq 2.0 \mu M$ in Table 1) extend to the right by 0.5 to 1.0 scale, including CCRF-CEM, K-562, MOLT-4, HCT-15, KM12, SNB-75, M14, CAKI-1, DU-145, and MDA-MB-435 cell lines from seven subpanels, suggesting that the responses against these cell lines are more selective than other lines. The selectivity between these ten most sensitive lines and the least sensitive lines (HCC-2998 and TK-10) is expressed by more than 50- or 100-fold. More interestingly, in the same cancer subpanel, a selectivity of more than 50-fold is observed between KM-12 and HCC-2998 (colon cancer), and between CAKI-1 and TK-10 (renal cancer), respectively. Very similar profiles are found at the TGI and LC_{50} levels, respectively.

The COMPARE analysis was further performed to elucidate the mechanism of action of protoneodioscin by comparison of the mean graphs of all compounds in the NCI's database (Paull et al. 1989; Weinstein et al. 1992). The ability of COMPARE to group compounds with similar mechanisms of action, molecular targets, or mechanisms of drug resistance, was well documented (Boyd and Paull, 1995; Bates et al. 1995; Monks et al. 1997; Weinstein et al. 1997). As a result, protoneodioscin was judged negative by the COMPARE, suggesting a potential novel mechanism of anticancer action involved.

In the follow-up animal studies, protoneodioscin was confirmed non-toxicity with the maximum tolerated dose (MTD) more than 600 mg/kg to mice, and therefore was highly evaluated by the NCI's Biological Evaluation Committee (BEC) (personal communication). This critical information was in coincidence with the fact that the rhizomes of *D. collettii* var. *hypoglauca* has been safely used in the treatment of cancer patients in China. Currently, protoneodioscin has been on the list for the *in vivo* hollow fiber assay in nude mice (Hollingshead et al. 1995).

In summary, the present paper reported the cytotoxicity of protoneodioscin (NSC-698789), a furostanol saponin isolated from the rhizomes of *D. collettii* var. *hypoglauca*, against human cancer cell lines *in vitro*. Our studies showed that protoneodioscin is cytotoxic against most cell lines from leukemia and solid tumors. Leukemia, CNS cancer, and prostate cancer are the most sensitive subpanels to protoneodioscin, while melanoma, ovarian cancer, and renal cancer are less sensitive. The preliminary animal studies showed that the maximum tolerant dose of protoneodioscin was 600 mg/kg to mice. Based on an analysis of the COMPARE software with protoneodioscin as a seed compound, no compounds in the NCI's database have similar cytotoxicity patterns to those of protoneodioscin, indicating a potentially novel mechanism of anticancer action involved.

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